

AI Roleplay Arena — Full Project Design

"Level up your Emotional Intelligence by battling AI personalities"

1. Core Concept

The site is framed as a **game arena**, not a chat app. The player picks an emotionally difficult "boss" character (an AI persona with a specific personality and emotional pattern), has a real-time conversation with them, and tries to de-escalate, empathize with, or communicate effectively — just like in real life. After the conversation, the AI generates an **EQ Report Card** scoring the player's performance, and successful players "unlock" harder bosses.

This works because it's simultaneously:

- **A real game** (bosses, levels, unlocks, scores) → visually and conceptually impressive
 - **Genuinely useful EI training** (practicing real conflict scenarios) → strong real-world justification
 - **100% LLM-powered** (persona roleplay + evaluation) → clean single-technology story for evaluators
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2. Full User Flow

Step 1 — Landing / Arena Home Screen

- Player sees an "Arena" screen with a row of **boss cards** (like a game character select screen)
- Each card shows: boss name, avatar/icon, difficulty stars, short teaser ("Never admits they're wrong. Deflects blame instantly.")
- Locked bosses are greyed out with a  icon and a hint like "Beat Passive-Aggressive Coworker to unlock"

Step 2 — Boss Selection

- Player clicks a boss card → a short **briefing screen** appears:
 - Boss backstory/context (e.g., "You're in a group project. Riya has missed two deadlines and gets defensive when confronted.")
 - The player's **objective** for this battle (e.g., "Get Riya to agree to a new deadline without the conversation turning into a fight")
 - A "Start Battle" button

Step 3 — The Battle (Live Chat)

- Standard chat UI, but styled like a game (health bar or "Tension Meter" instead of plain chat)
- Player types responses; the AI (in-character) replies realistically — it can get defensive, calm down, escalate, or de-escalate depending on how well the player communicates
- A live **Tension Meter** at the top visually reacts after each message (goes up if player is harsh/dismissive, goes down if player is empathetic/calm) — this is the "health bar" game feel
- Battle ends after a fixed number of turns (e.g., 8-10 exchanges) OR when Tension Meter hits 0 (resolved) or max (blown up/failed)

Step 4 — EQ Report Card (Results Screen)

- AI generates a post-battle summary styled like a game results screen:
 - **Outcome banner:** "Conflict Resolved!" / "Partial Success" / "Conversation Escalated"
 - **EQ stats** broken into categories (e.g., Empathy: 7/10, Self-Regulation: 8/10, Clarity: 6/10, Active Listening: 9/10)
 - **Highlight moments:** 1-2 specific quotes from the player's messages with AI commentary ("When you said 'I understand you're overwhelmed,' Riya's defensiveness dropped — great empathy move.")
 - **XP earned** and whether a new boss was unlocked

Step 5 — Progression Screen

- Player returns to Arena Home; newly unlocked boss card animates/highlights
- A simple **profile/progress page** shows total battles played, average EQ score, and a growth chart over time (radar chart of Empathy/Regulation/Clarity/Listening across all past battles)

3. The "Boss Roster" (Suggested for MVP — 4 bosses)

Boss	Personality Pattern	Difficulty	Skill Trained
Passive-Aggressive Coworker	Deflects, sarcastic, avoids direct blame	★ Easy	Direct, calm confrontation

Boss	Personality Pattern	Difficulty	Skill Trained
Overreacting Sibling	Escalates quickly, emotional, feels unheard	☆☆ Medium	Active listening, validation
Anxious Friend	Spirals, seeks reassurance, catastrophizes	☆☆ Medium	Patience, grounding language
Defensive Parent/Boss (authority figure)	Dismissive of feelings, shuts down conversation	☆☆☆ Hard	Assertiveness + tact combined

(Start with these 4 for the demo; more can be "coming soon" locked cards for visual effect even if not built.)

4. How the AI Actually Works (Technical Logic)

You'll use **two distinct LLM calls per battle role**, both via the same API but with very different system prompts:

A) In-Character Roleplay Call (during the chat)

Every time the player sends a message, you call the LLM with:

- A **system prompt** defining the boss's personality, backstory, and emotional rules
- The **full conversation history** so far (for context/memory)
- An instruction to reply **ONLY** in-character, in 1–3 sentences (keeps it snappy, game-like)

Example system prompt (Passive-Aggressive Coworker):

You are roleplaying as "Riya," a coworker on a group project who has missed two deadlines. Personality: passive-aggressive, deflects blame with sarcasm, avoids direct confrontation, gets defensive if criticized bluntly, but responds well to calm, non-accusatory, empathetic language.

Rules:

- Stay fully in character as Riya. Never break character or mention you are an AI.
- Keep replies short (1-3 sentences), realistic, and emotionally reactive.
- If the player is harsh, blunt, or accusatory, become more defensive/sarcastic.
- If the player is calm, specific, and empathetic, gradually soften and become more cooperative.
- Track the emotional trajectory across the conversation – don't reset emotion each turn.

Alongside the reply, ask the model to also output a hidden **tension_delta** value (e.g., -2 to +2) representing how that exchange affected the Tension Meter — this can be requested as structured JSON so your frontend can animate the meter.

B) Evaluation Call (after the battle ends)

Once the battle is over, send the **entire conversation transcript** to the LLM again with a *different* system prompt that asks it to act as an EI coach/evaluator:

You are an Emotional Intelligence coach. Below is a roleplay conversation between a user (practicing EI skills) and an AI playing a difficult character named Riya. Evaluate the USER's messages only.

Return a JSON object with:

- empathy_score (1-10)
- self_regulation_score (1-10)
- clarity_score (1-10)
- active_listening_score (1-10)
- outcome ("Resolved" / "Partial" / "Escalated")
- two highlight_moments: each with a short quote from the user and a one-sentence explanation of why it worked or didn't work
- one_tip_for_improvement (short, encouraging, specific)

This structured JSON output is what powers your Report Card screen.

5. Main Features List

- 🎮 Boss-select arena screen with unlock progression
 - 💬 Real-time in-character AI chat battles
 - 📊 Live "Tension Meter" that reacts to message tone each turn
 - 🏆 AI-generated EQ Report Card after every battle (scores + highlights + tip)
 - 🔒 Unlock system (beat easier bosses to access harder ones)
 - 📈 Personal growth dashboard (radar chart across all past battles)
 - 🎭 4 distinct AI personas with different emotional patterns
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6. Recommended Tech Stack

Layer	Recommendation
Frontend	React (or plain HTML/CSS/JS if time is tight)
Styling	Tailwind CSS for quick, clean game-like UI
AI	LLM API (Claude/OpenAI/Gemini) — two prompt types: roleplay + evaluation
Backend	Node.js + Express (to keep API key secure, manage conversation state)
Database	Firebase Firestore (store user progress, unlocked bosses, past battle scores)
Charts	Chart.js or Recharts for the radar/progress chart
Auth (optional)	Firebase Auth (simple email/anon login) so progress persists

7. Suggested Build Order (3-Week Plan)

Week 1 — Core chat engine

- Set up backend endpoint that takes boss ID + message history → returns in-character reply
- Build basic chat UI, hardcode 1 boss to test end-to-end flow
- Get the tension_delta JSON output working reliably

Week 2 — Game layer

- Build Arena home screen with boss cards + lock/unlock logic
- Add Tension Meter animation
- Add remaining 3 boss personas (just new system prompts — fast to add)
- Implement the evaluation call → Report Card screen

Week 3 — Polish + progression

- Firebase integration: save battle history, unlock state, scores
 - Build the growth dashboard (radar chart across battles)
 - UI polish: sound effects/animations optional but big "wow factor" for evaluation
 - Test edge cases (player being rude, giving up mid-battle, very short replies)
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8. Why This Scores Well in Evaluation

- **Visually obvious AI use** — faculty can literally watch the AI stay in character and react differently to good vs. bad responses, live
 - **Two distinct AI capabilities shown** — persona roleplay AND structured evaluation/scoring, which shows deeper understanding than a single chatbot call
 - **Game mechanics justify the complexity** — unlocks, meters, and report cards make the UI naturally more polished-looking than a typical form-based project
 - **Directly maps to EI syllabus concepts** — empathy, self-regulation, active listening, communication are literally the scored categories
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9. Future Scope (for the "future work" slide)

- Voice-based battles (speech-to-text input, AI reads boss lines aloud with tone)
- Difficulty auto-adjustment based on player's ongoing performance
- Multiplayer mode: two real users practice mediating a conflict with AI as judge
- Teacher/classroom mode: aggregated anonymized class EQ growth stats
- More bosses covering workplace, family, and romantic relationship scenarios